

Economic History II

Bureaucracy and patronage

FGV EPGE

Outline of the class

- How bureaucrats are appointed, and what the method of appointment does to them.
- A principal-agent framework for bureaucracy: Besley, Burgess, Khan and Xu (2021).
- Two papers by Guo Xu read through it: Xu (2018) on British colonial governors, Voth and Xu (2022) on the Royal Navy.
- Both papers use the same measure of connection, in the same state, and find opposite effects.

Government and development (Lucas, 1988)

"Is there some action a government of India could take that would lead the Indian economy to grow like Indonesia's or Egypt's? If so, what, exactly? If not, what is it about the 'nature of India' that makes it so? The consequences for human welfare involved in questions like these are simply staggering: Once one starts to think about them, it is hard to think about anything else."

Robert Lucas, 1988

Where bureaucracy enters

- Government action is carried out by the **bureaucrat**, not by the politician.
- Taxing, enforcing law, building infrastructure, delivering services: all of it runs through paid officials.
- Weber's ideal type: rule-driven, professional, hierarchical, security of tenure in exchange for a sense of duty, "fealty to the purpose of the office".
- The rival tradition: bureaucracy as sclerotic, self-interested, captured by the firms it regulates.

A brief history of bureaucracy

- The first bureaucratic tasks were collecting taxes and enforcing laws. Colonial empires then needed administrators abroad as well as at home.
- China: a professional cadre of *mandarins*, recruited through competitive examinations.
- England: after the Glorious Revolution, a meritocratic excise bureaucracy built to pay for naval expansion.
- Northcote-Trevelyan report (Britain, 1854): competitive entry exams, transparent rules of promotion, a permanent civil service. It also shaped the administration of Britain's colonies.
- Pendleton Act (United States, 1883): a first restraint on the spoils system in federal appointments.

Assassination of James A. Garfield

🌐 22 languages ▾

Article [Talk](#)

Read [Edit](#) [View history](#)

From Wikipedia, the free encyclopedia

Coordinates: 🌐 38°53′31″N 77°01′13″W

On July 2, 1881, at 9:30 AM, [James A. Garfield](#), the 20th [president of the United States](#), was shot at the [Baltimore and Potomac Railroad Station](#) in [Washington, D.C.](#), less than four months into his term as president.

The assassin was [Charles J. Guiteau](#), a mentally ill and delusional [office seeker](#), who had distributed copies of a speech he wrote aimed at promoting Garfield in the [1880 United States presidential election](#). Guiteau believed his campaigning had been vital to Garfield's eventual victory, and that Garfield owed him a diplomatic post in Europe for his assistance. After months of failed attempts to solicit such a reward from the Garfield administration, he purchased a revolver and began stalking Garfield with the goal of assassinating him, hoping that if he succeeded in doing so, Garfield's vice president [Chester A. Arthur](#) would give him the job he wanted upon assuming the presidency.

Assassination of James A. Garfield



President Garfield with [James G. Blaine](#) after being shot by [Charles J. Guiteau](#)^{[1][2]}

Location 🌐 38°53′31″N 77°01′13″W
[Baltimore and Potomac Railroad Station](#).

Pendleton Civil Service Reform Act

 7 languages 

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From Wikipedia, the free encyclopedia

The **Pendleton Civil Service Reform Act** is a [United States federal law](#) passed by the [47th United States Congress](#) and signed into law by President [Chester A. Arthur](#) on January 16, 1883. The act mandates that most positions within the [federal government](#) should be awarded on the basis of merit instead of political patronage.

By late 1826, American politics operated on the [spoils system](#), a political patronage practice in which officeholders awarded their allies with government jobs in return for financial and political support. Proponents of the spoils system were successful at blocking meaningful civil service reform until [the assassination](#) of President [James A. Garfield](#) in 1881. The 47th Congress passed the Pendleton Civil Service Reform Act during its [lame duck session](#) and President [Chester A. Arthur](#), himself a former spoilsman, signed the bill into law.

The Pendleton Civil Service Reform Act provided for the selection of some government

Pendleton Civil Service Reform Act

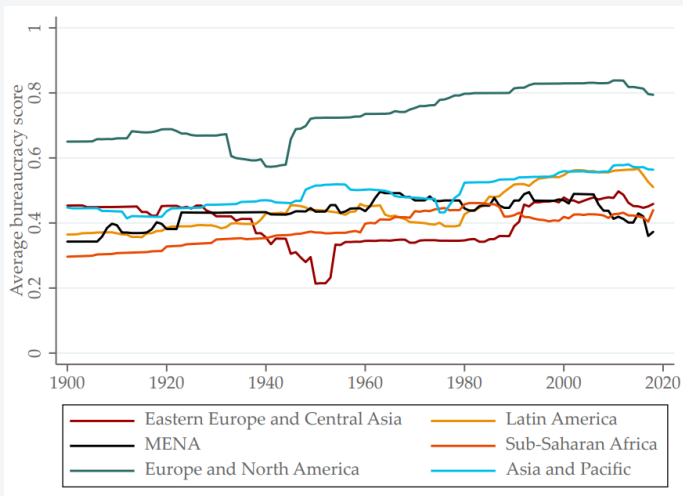


Long title	An act to regulate and improve the civil service of the United States.
Nicknames	Pendleton Act
Enacted by	the 47th United States Congress
Effective	January 16, 1883

Three facts before the theory

- Besley et al. score countries on two dimensions: meritocratic recruitment, and rigorous and impartial administration.
- **Persistence**: differences across countries account for 73% of the variation. Ranks barely move over a century.
- **Level**: no country above the high-income threshold scores below the 42nd percentile of bureaucratic quality.
- **Growth**: within a country, improvements in the score track higher income. Correlations only; the direction of causality is open.

Persistence: bureaucratic quality by region since 1900



Bureaucracy as a principal-agent problem

- A principal (the state, or the minister who speaks for it) delegates a task to an **agent** (the bureaucrat).
- The agent has objectives of his own, and information the principal does not have.
- The principal does not observe how hard the agent works.
- Unlike politicians, bureaucrats are picked by a superior, not elected.

Type and effort

- Performance depends on three things: the bureaucrat's **type** (talent, character), his effort, and the difficulty of the task.
- His type is fixed; his effort is chosen.
- The principal observes neither directly, which creates two separate problems.

Selection and incentives

- *Selection*: put the right type in the post. Who enters, who is promoted.
- *Incentives*: get effort out of the man already in post. Monitoring, promotion, prestige, dismissal.
- Explicit performance pay is rare in the public sector, and firing costs are high, so promotion carries most of the weight.
- Xu (2018) finds effects through both; Voth and Xu (2022) only through selection.

Measuring performance

- Any incentive scheme needs a mapping from output to reward.
- Bureaucratic goals are multiple, non-market and often not even verifiable after the fact.
- Tractable cases are frontline: tax collectors, teachers, nurses. Senior officials who write rules are much harder.
- One approach, taken from the literature on **CEOs**, is to map one senior official onto one aggregate outcome.
- Besley et al. cite Xu (2018) as an example: governors mapped onto their colony's revenue.

Rules or discretion

- Appointment by rule: entry exams, seniority, published criteria, fixed in advance.
- Appointment at the discretion of a superior: **patronage**.
- A rule limits favoritism, but it cannot use what the appointer privately knows about the candidate.
- Northcote-Trevelyan (1854) in Britain and the Pendleton Act (1883) in the United States both pushed toward rules.

What the literature usually finds

- Politically aligned appointees in Brazil are less qualified (Colonnelli, Prem and Teso).
- Managers who override hiring tests get hires who quit sooner.
- Voth and Xu, on the state of the field: it treats “virtually any kind of discretion as a guarantee for corruption and inefficiency”.
- Xu (2018) fits this view. Voth and Xu (2022) find the opposite.

Patronage in the British Empire

Xu (2018) and Voth and Xu (2022)

Xu (2018): the question

- The object is the Colonial Office, the bureaucracy that administered Britain's overseas possessions.
- Each colony was headed by a governor, appointed by the Secretary of State for the Colonies, a politician in London.
- Until 1930 that appointment was purely discretionary: it was **patronage**.
- Question: what did a personal connection between governor and minister do to the governor's promotion, and to his performance?

The Costs of Patronage: Evidence from the British Empire

By GUO XU*

I combine newly digitized personnel and public finance data from the British colonial administration 1854-1966 to study how patronage affects the promotion and incentives of governors. Governors are more likely to be promoted to higher salaried colonies when connected to their superior during the period of patronage. Once allocated, they provide more tax exemptions, raise less revenue, and invest less. The promotion and performance gaps disappear after the abolition of patronage appointments. Patronage therefore distorts the allocation of public sector positions and reduces the incentives of favored bureaucrats to perform.

JEL: O1, M51, D73



FIGURE 1. TERRITORIES ADMINISTERED BY THE COLONIAL OFFICE - 1905

The Colonial Office

- Founded in 1854, absorbed into the Foreign Office in 1966.
- At its peak it administered close to a fifth of the world's land mass.
- 70 colonies in the data. India is not among them: it had its own department, the India Office.
- Governorships were held to be *“proper objects for the exercise of patronage by the Secretary of State for the Colonies.”*

The governor as agent

- Appointed for a fixed term of six years, reporting directly to the Secretary of State.
- Duties codified in the Colonial Rules and Regulations: public finance, legislation, colonial appointments, security, pardons.
- In practice the **man on the spot**: “the distance between the centre and the periphery required a policy of trust”.
- London’s only real levers: “Their original act is sending a good governor, and their check is dismissing him.”

Why revenue measures a governor

- Under the **revenue imperative**, each colony had to finance its own public services out of taxes it raised.
- Jeffries: “the colonies were expected to pay their way” .
- The Colonial Office itself judged governors by the revenue they raised.
- Xu checks it: promotions at the end of the six-year term did track past revenue performance.

Data

- Over **3,000 volumes** of personnel and public finance records, digitized from the National Archives and Commonwealth libraries.
- Colonial Office Lists: postings, careers, salaries.
- Blue Books: annual revenue, expenditure, population, trade, prices, on standardized forms the governors had to file.
- The universe: 456 governors, 70 colonies, 37 Secretaries of State, 1854 to 1966.

COMPARATIVE YEARLY STATE.

Specify each separate Tax or Duty.	Amount collected in the Year 1845 in Pounds Sterling.			Amount collected in the Year 1846 in Pounds Sterling.		
	£	s.	d.	£	s.	d.
Customs Revenue	69,412	5	7	69,756	9	4
Light Dues.	2,358	12	7	2,437	10	3
Rents of Crown Lands Proceeds of Crown Lands and Rents Redeemed	821	14	3	721	1	7
Licence Fund Fines and Forfeitures.	789	1	5	1,152	8	2

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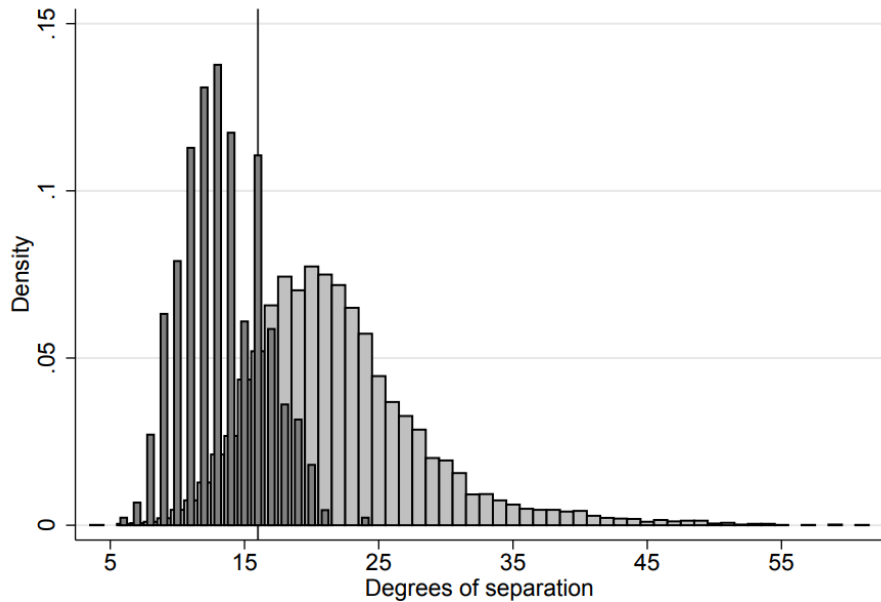
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Measuring connection

- Genealogy (The Peerage) plus biographical directories (DeBrett's, Who's Who).
- **Connected** if any of four holds: shared ancestry, both in the peerage, both at Eton, both at Oxford or both at Cambridge.
- Shared ancestry means within 16 degrees of separation, roughly eighth cousins. That covers 25% of governors.
- All four traits are fixed before the career starts, so they cannot be produced by the governor's own performance.



Dark grey = Observed ties; Light grey = Randomly drawn

Reading the distribution

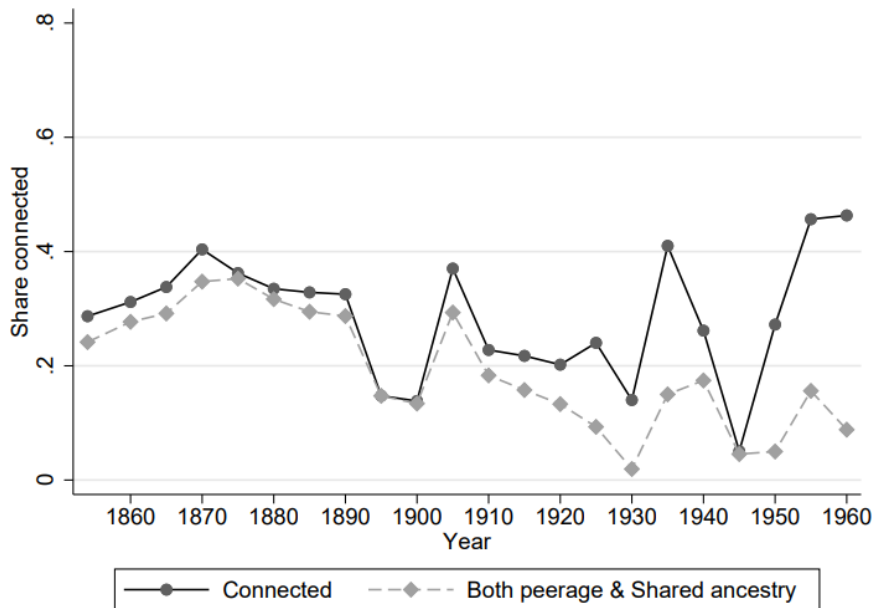
- Dark: observed governor-minister pairs. Light: pairs drawn at random from the peerage.
- Observed ties sit well to the left. Of 1,000 random peerage pairs, only 10% are closer than 16 degrees.
- The cutoff is a trade-off: closer ties are more likely to be real ties, but leave fewer governors who change status.
- 94% of Secretaries of State are matched in the genealogy, only 34% of governors, who came from less elite circles.

First source of variation: ministers turn over

- The Secretary of State changed on average every three years, through cabinet reshuffles and elections in London.
- Every change reshuffles connections to all serving governors at once, for reasons having nothing to do with any colony.
- So the same governor is connected in one year and not in the next.
- About **21%** of the 456 governors switch status during their career.

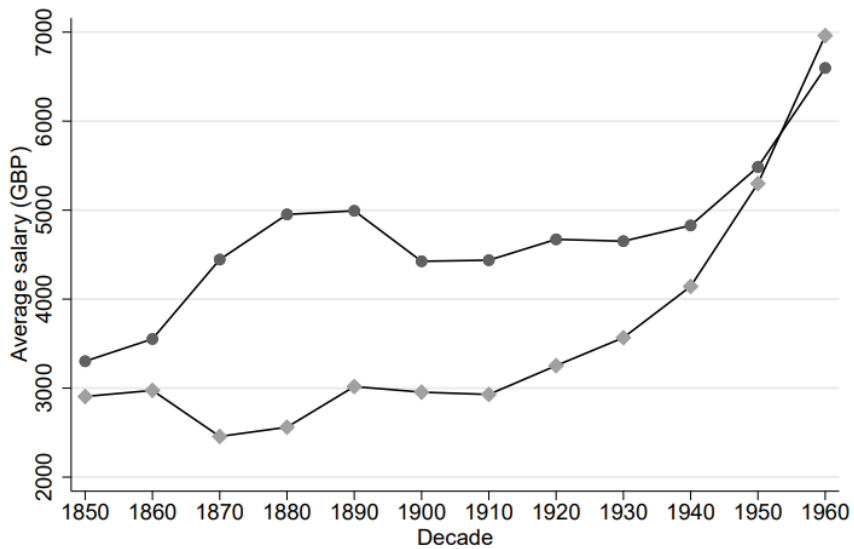
Second source of variation: Warren Fisher, 1930

- Before 1930 the Secretary of State selected governors himself, through his private secretary of appointments.
- The 1930 report: the “system is open to criticism first and foremost as being at any rate in theory, a system of patronage”.
- It handed recruitment to the Colonial Service Appointments Board, nominated by the independent civil service commission.
- Contemporaries called it the **“Magna Carta of the Colonial Service”**.



Connectedness over time

- Roughly a quarter to a third of serving governors are connected at any date.
- The composition shifts: family and peerage ties fall, school and university ties rise.
- The overall share is flat enough that 1930 is not itself a break in who is connected.



Salaries, connected and unconnected

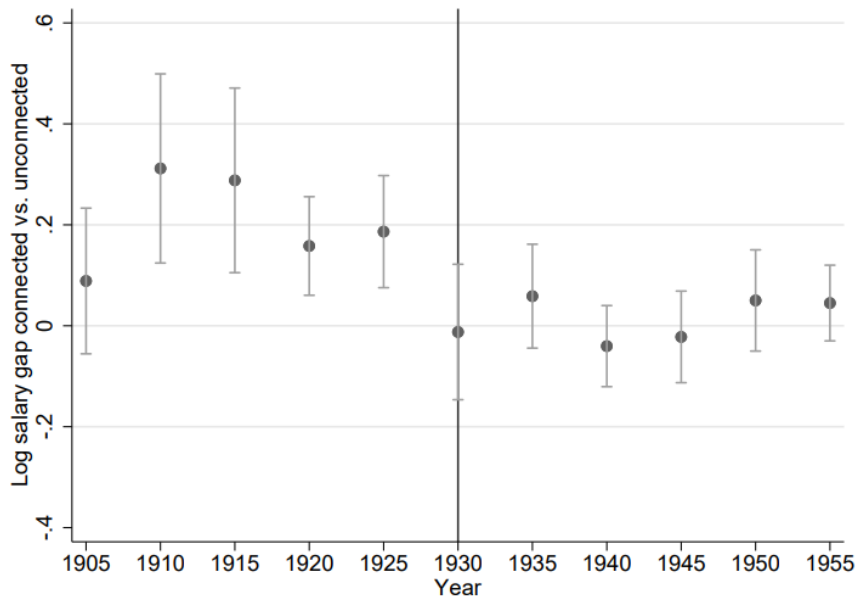
- Connected governors are paid more for most of the period, with the widest gap in the late nineteenth century.
- The gap closes after 1930.
- These are raw averages: connected men might simply be abler, and pay might compress as decolonization sets in.
- The rest of the paper is about ruling those two readings out.

Result 1: the same governor, better paid

- Follow one governor as his connection status changes: about **10% higher salary** when connected.
- Salaries attach to posts, not to men. Within the same colony the premium is close to zero.
- So the premium is a transfer: connected governors are moved to larger, higher-paying colonies.
- Worth roughly half of what an entire extra colony of experience is worth.

What the better posts look like

- Higher-paid governorships are larger and richer colonies, by the revenue they were raising when the Colonial Office took them on.
- They are not more tropical and not further from London, so this is not hardship pay.
- If anything, colonies with higher settler mortality paid less.
- Connection moved the governor to more **desirable** jobs, holding his type fixed.

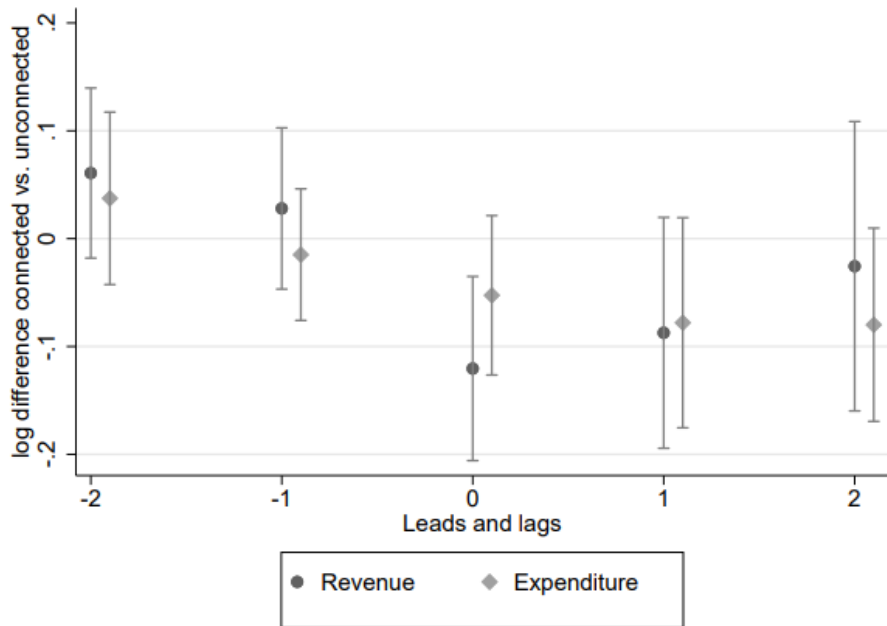


Before and after 1930

- The salary gap between connected and unconnected, in five-year bins around the reform.
- Positive and large through the 1910s and 1920s: **12.7%** before 1930.
- After 1930, indistinguishable from zero.
- Holds when the war years and the Great Depression are dropped, and when connection is allowed its own trend.

Result 2: performance

- Now hold the post fixed: same governor, same colony, connection switching under him.
- When connected he raises about **4% less revenue** a year.
- The fall is in customs, which was where the bulk of colonial revenue came from.
- Spending falls with it: about 8.9% less on revenue collection services, 10.7% less on public works.



Timing of the drop

- Nothing in the two years before a governor becomes connected, so this is not a trend that was already running.
- The gap opens in the year the connection arrives and persists afterwards.
- Revenue and expenditure move together, since colonies had to finance their own spending.

The channel: tax exemptions

- Governors legislated on tax by ordinance and proclamation, and had to report each change to London.
- Connected governors issue more ordinances, and the increase is concentrated in tax and customs.
- Xu read and coded 405 governor-years of laws: connected governors are more likely to add an **exemption** to the import tariff schedule.
- Import duties fell on the settler elite, “mainly levied on luxury items such as spirits, beer, wine, tobacco, fire-arms, gunpowder and manufactured cloth”.

FIGURE A7. SAMPLE OF TRADE TAX EXEMPTION LAWS (BLUE BOOK)

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Act of Customs which levied.

Specification of Taxes, Duties, &c.

EXEMPTIONS.—Continued.

<i>Machinery.</i> —Under Ordinance No. 2 of 1859, and Minute of 8th June, 1865.	-	-	-
All descriptions of Marine, Locomotive, Stationary, and Portable Steam Engines, their Boilers, and Gear, whole, or in parts	-	-	-
Machinery adapted to Windmills	-	-	-
Every description of Machinery for lifting, forcing, and conducting water	-	-	-
Machinery for the manufacture of Sugar, Hydraulic, and Screw presses	-	-	-
Cranes, Crab Winches, and Screw Jacks	-	-	-
Pulpers, Peelers, and Winnowing Machines, whole, or in parts	-	-	-
Tile, Brick, and Pipe making machines	-	-	-
Printing and Lithographic Presses	-	-	-
Fibre, Cotton Carding, Weaving and Spinning Machines	-	-	-
Nasmyth's Hammers	-	-	-
Lathes	-	-	-
Punching, Drilling, Shearing, Planing, Sawing, and Screw making Machines	-	-	-
All materials and plant imported for the construction of the Railway between Colombo and Kandy, certified by the Chief Resident Engineer as required for that purpose	-	-	-
Manures	-	-	-
Paper	-	-	-
Pepper, Black	-	-	-
Regimental Clothing, Necessaries, and Accoutrements, imported for the use of Her Majesty's Land and Sea Forces	-	-	-
Seeds intended for Agricultural and Horticultural purposes, including Plants	-	-	-
Specimens illustrative of Natural History	-	-	-
Tanks (Iron)	-	-	-
Whale Oil	-	-	-

Free.

Ordnances No. 18 of 1852,
No. 9 of 1853, No. 2 of
1856, No. 2 of 1859, and
No. 3 of 1862.

EXPORT DUTIES.

DUTIES of CUSTOMS payable on Goods, Wares, and Merchandize, being the Growth, Produce, or Manufacture of the Island of Ceylon, exported to parts beyond seas.

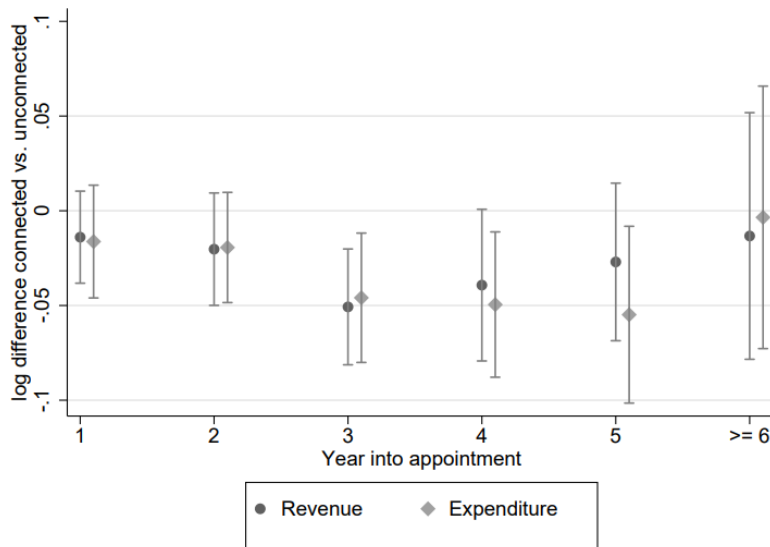
ARTICLES.			RATE OF DUTY.		
			£	s.	d.
Arrecanuts	-	-	-	-	-
Cinnamon	-	-	-	-	-
Coffee	-	-	-	-	-
Cable Yarn, Fibre, Rope, and Junk	-	-	-	-	-
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Note: Sample of customs tax exemptions laws from the 1869 Ceylon Blue Book.

Reading the schedule

- Ceylon Blue Book, 1869: machinery, printing presses, railway plant, all entering free.
- Exemptions were recorded systematically, and each one lowers trade revenue directly.
- Raising taxes in a colony required legitimacy, and governors balanced local elites against the directives from London.
- Import duties were a frequent source of colonial conflict; the Boston Tea Party was a protest against one.

FIGURE A6. FISCAL PERFORMANCE AND CONNECTEDNESS BY YEAR INTO APPOINTMENT



Where in the term the gap appears

- Secretaries of State last about three years; a governor's term lasts six.
- A governor connected in year one is unlikely to face that same minister when the next posting is decided.
- The revenue gap shows up in the later years of the term, when the connected minister is the one who will decide.
- Consistent with connection acting as a **substitute** for performance in the promotion decision.

Other measures of performance

- Riots reported in UK newspapers: colonies under connected governors are 3.8 percentage points more likely to appear.
- Parliamentary debates: those colonies are mentioned with more negative sentiment.
- Honors: connected governors are 3.1 percentage points less likely to receive the Knight Grand Cross, against a base rate of 2%.
- All three gaps close after 1930.

Xu (2018) in the framework

- In the Colonial Office, discretion worked as **favoritism**.
- Selection: the best posts went to the connected, holding ability fixed.
- Incentives: with the next promotion already secured by the connection, effort falls.
- The paper does not claim connected governors were bad men, only that connection and performance were substitutes.

The Royal Navy

Voth and Xu (2022): promotion, performance and patronage

Voth and Xu (2022): the question

- A second paper by Xu, with Voth, on an earlier period: the Royal Navy, 1690 to 1849.
- The Admiralty had formal discretion over officer appointments, what contemporaries themselves called patronage.
- The decisive step in a career was promotion to **post-captain**: independent command, up to 950 men, no supervision at sea.
- Question: did that discretion pick better captains, or protect relatives?

Discretion and Destruction: Promotions, Performance, and Patronage in the Royal Navy'

Joachim Voth
University of Zurich
& CEPR

Guo Xu
University of California,
Berkeley Haas

February 24, 2022

Abstract

How effective is discretion for selecting leaders? We collect new data on the performance of more than 5,800 naval officers serving in the 18th century Royal Navy. Officers fast-tracked for promotion at the discretion of the Admiralty performed markedly better than those “passed over” – the Navy picked future winners. Selection was particularly effective for officers connected to the Admiralty through family ties. Instead of favoritism, we find that family ties and discretion allowed the effective use of private information. Keeping out the “wrong” connected officers while fast-tracking the high-performing ones was a key driver of superior selection.

The setting

- The Navy grew from 147 to 752 larger ships between 1690 and 1810; over 160,000 men at the Napoleonic peak.
- No European navy had a technological edge, yet outcomes were lopsided: in the Napoleonic Wars Britain lost 166 ships and destroyed or took 1,201, a ratio of 7 to 1.
- Naval commissions could not be purchased. Voth and Xu argue that captains differed mainly in **character**.
- Regulations punished a lack of fighting spirit because there was “a real problem of cowardice”.

What the Admiralty was trying to predict

- Appointing a coward to independent command was to court disaster, and the decision was effectively irreversible.
- Rodger: “For a young officer who hoped for command, or for post-rank, nothing was more swiftly effective than to take an enemy ship of equal or greater force.”
- British ships were expected to fight above their weight: “British commanders were expected to defeat enemy forces much stronger than their own”.
- At the time of promotion, the Admiralty had little hard information on how a lieutenant would perform in command.

In North America.

under the command of the R^t Hon^{ble} Lord Viscount Howe;

3 Eagle	45 520	F. J. P. Moore H. P. Duncan	7 th 7 th 1/2	7 th 7 th 1/2
Autent	45 515	P. A. Spangler Geo. Stippel	7 th 7 th 1/2	7 th 7 th 1/2
Strident	45 500	Tom. Platt H. P. Moore H. P. Moore John Galt	7 th 7 th 7 th 7 th	7 th 7 th 7 th 7 th
Conewet	45 520	Geo. Curry S. P. Moore S. P. Moore S. P. Moore S. P. Moore	7 th 7 th 7 th 7 th 7 th	7 th 7 th 7 th 7 th 7 th

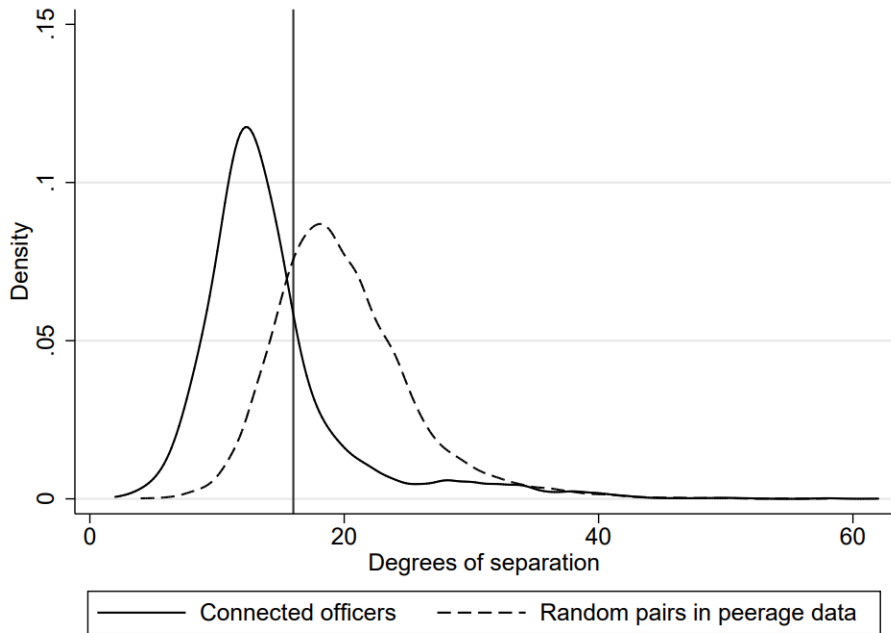
Ship	Yr	Commander	Lieutenant	What Crew?	What last died	What suffered from England	Disposition
3 Phoenix	63 500	Wm Gifford	Paul Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
4 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
5 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
6 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
7 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
8 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
9 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
10 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
11 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
12 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
13 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
14 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
15 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
16 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
17 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
18 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
19 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				
20 Phoenix	63 500	Wm Gifford	John Butler	776			2 Mar
			Wm Brown				
			Wm Wadsworth				
			A				

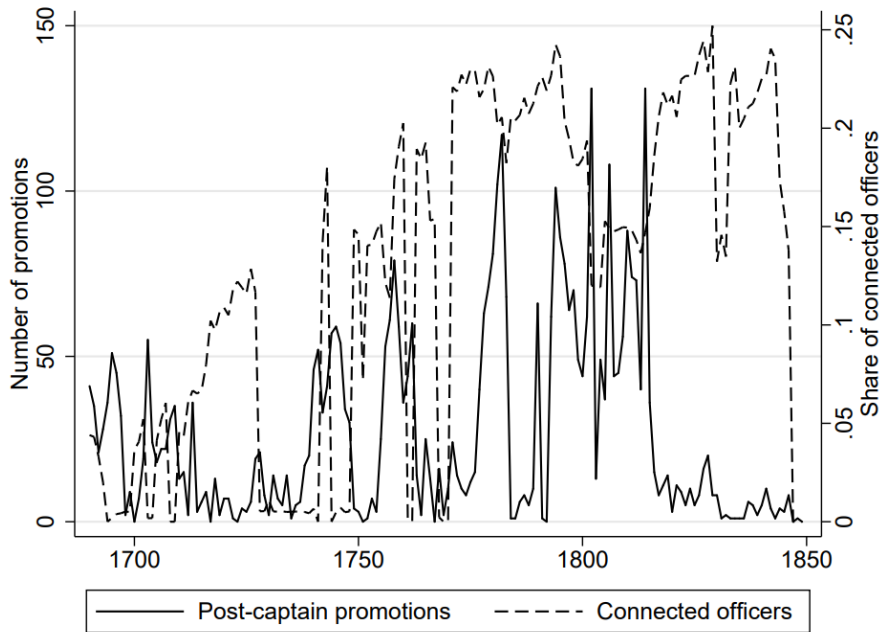
Data

- Threedecks, a database of navies in the Age of Sail, plus newly digitized Admiralty List Books from the National Archives.
- **5,848 officers**, 3,904 ships, 56 stations, 1690 to 1849.
- 2,782 promotion decisions taken by 49 Admiralty Boards.
- For every decision the whole choice set is observed: who was promoted, and who was passed over.

Measuring performance and connection

- Performance is **victories**: enemy ships captured or sunk. Logged on the ship and widely reported at the time.
- On average an officer could claim one victory in a career.
- Connection is a family tie to the two men at the top, the First Lord of the Admiralty and the Admiral of the Fleet.
- Same genealogy and same 16-degree cutoff as Xu (2018). 562 of the 5,848 officers were ever connected.





Promotions and connections over time

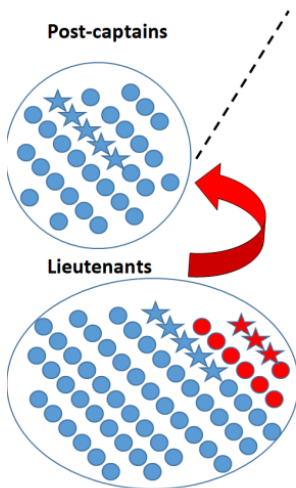
- Promotions come in wartime spikes: the Navy makes captains when it needs them.
- The connected share rises through the eighteenth century and peaks in the French Revolutionary Wars. Across all years it averages 11%.
- It then falls sharply under John Jervis, a career admiral from a middling background, who set out to curtail “influence” in appointments.

The identification problem

- To judge a promotion you need the performance of the man who was not chosen, and normally you never see it.
- In the Navy it is possible, because discretion mostly affected the timing of promotion to captain, and 53% of lieutenants were eventually promoted.
- So compare the fast-tracked with the **passed over**, once the passed over finally get their command.
- Then compare that gap for connected and for unconnected officers.

Figure 2: Construction of counterfactual - illustration

Stock of officers



Choice year 1700:

Officer A - Connected

Officer B - Connected

Officer C - Unconnected

Officer D - Unconnected

Officer B counterfactual - Connected

Officer B counterfactual - Unconnected

Unconnected

● Promoted

● Active

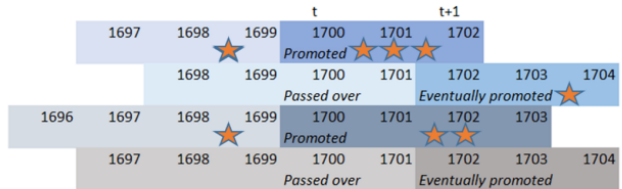
Connected

★ Promoted

★ Active

Annual performance and promotions

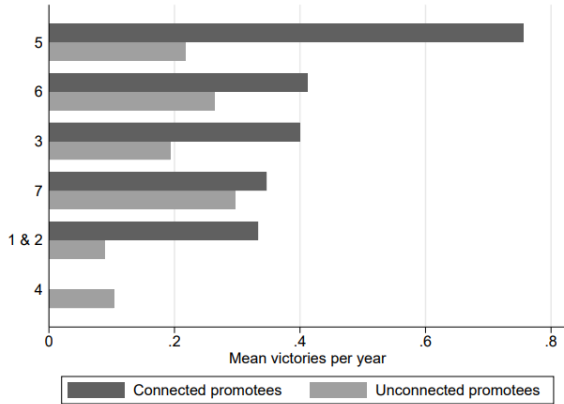
★ enemy capture



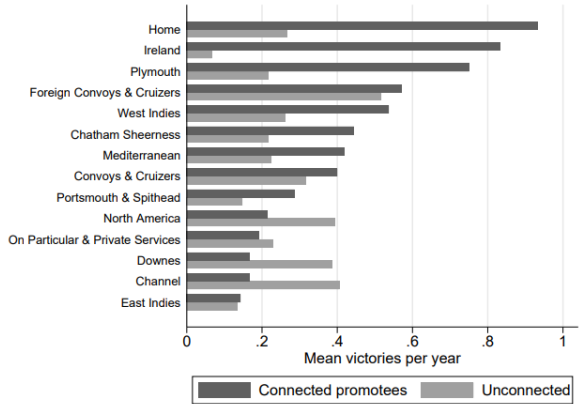
Reading the design

- Choice year 1700: A and C are promoted, B and D are passed over.
- B and D make captain in 1702. Their capture rate as captains stands in for what they would have done from 1700.
- Differencing connected against unconnected removes whatever being promoted late does to a man on its own.
- The data support that step: being passed over has no different effect on connected and unconnected officers.

Figure 1: Average performance of post-captains by connectedness at time of promotion



(a) By ship rate (class)



(b) By station/task

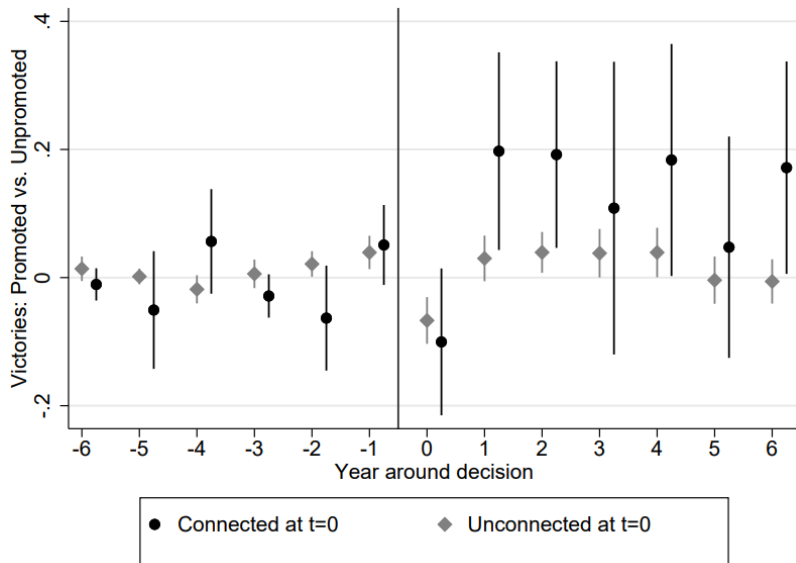
Raw performance

- Mean victories per year, by class of ship and by station.
- Small fast ships sent on independent cruises capture most; large battleships on blockade duty capture least.
- Connected promotees are ahead in almost every class, and on 10 of the 14 stations.
- That already argues against the connected simply being handed the easy assignments.

Result: the fast-tracked fight better

- Officers promoted straight away capture about **11%** more per year than those passed over.
- The gap is concentrated among the connected: about a quarter above the average rate.
- Connected officers who were passed over do not outperform, so connection alone does not explain the gap.
- Holds comparing officers on the same station, the same ship and the same crew, with ship age, refit and gun count controlled.

Figure 4: Performance of promoted vs. unpromoted officers around decision year, by connectedness



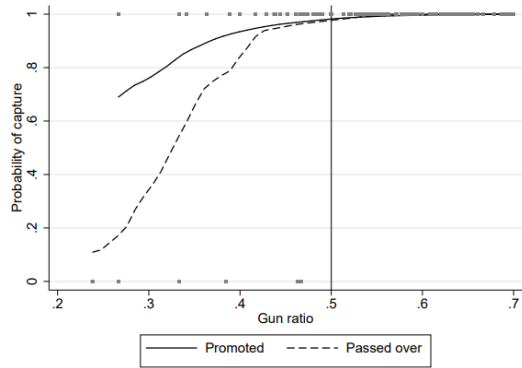
Around the decision year

- Before the decision, promoted and passed-over officers capture at the same rate, in both groups.
- After promotion the connected fast-trackees pull away; the unconnected barely move.
- Survives matching each promotee to an officer with the same decision year, the same fighting record, similar tenure and similar command experience.
- The admirals' choices used information that was not in the service record.

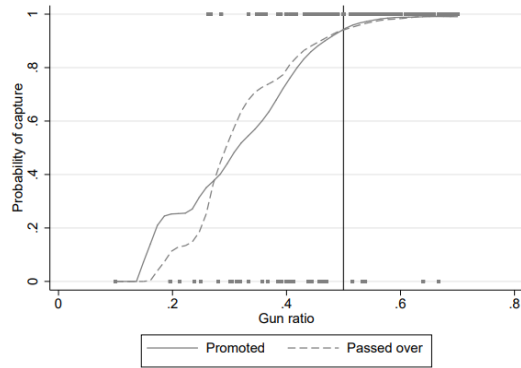
Assignment, or ability?

- Connected promotees are no more likely to be sent to the high-capture stations.
- Ship age, time since refit, crew quality and the ship's own track record are all unrelated to being a connected promotee.
- Degree of **discretion** matters: overseas, where station commanders made appointments the Admiralty found costly to overturn, the advantage disappears.
- The advantage is concentrated in home waters, where the Admiralty had full discretion.

Figure 3: Probability of victory by gun ratio for promoted vs. passed over officers, broken down by connectedness



(a) Connected to Admiralty



(b) Unconnected to Admiralty

Fighting at a disadvantage

- Probability of taking the enemy ship, plotted against relative firepower (own guns over total guns in the action).
- Evenly matched, a British ship won almost always. Outgunned two to one, still roughly an even chance.
- Outgunned two to one, connected fast-trackees won **63%** of the time; connected officers who had been passed over, 12 to 18%.
- Among the unconnected the two curves are close together.

Single ship duels

- 172 chance encounters on the high seas: 115 enemy ships taken or sunk, 16 British ships lost, 41 inconclusive.
- Connected fast-trackees are 24 percentage points more likely to win, against a base rate of 65.6%.
- Controlling for the gun ratio barely moves that.
- Their engagements are almost never indecisive, against nearly a quarter under unconnected captains.

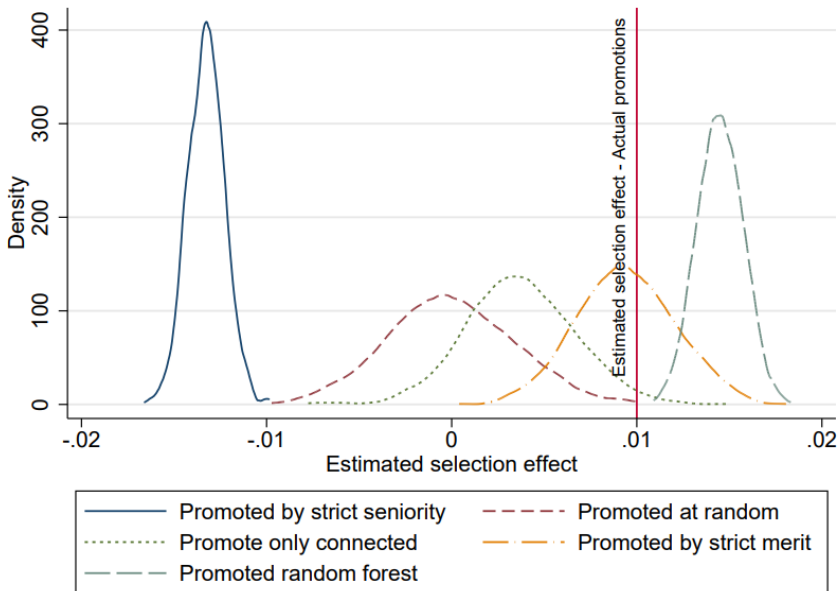
Fighting spirit

- Breaking off an action without heavy casualties was dishonorable and could end in a court martial.
- The evidence suggests the admirals were judging an officer's willingness to fight an action to the end.
- Closer kin are picked better: doubling the degrees of separation costs about 7.2% of the annual victory rate.
- Relatives who waited longer for promotion performed worse once promoted. There is no such pattern among the unconnected.

What the rules would have done

- Rerun the choices under alternative rules, holding the number of promotions each year at the actual figure.
- Strict seniority: worse than promoting at random, because it holds back young officers who have already done well.
- Promote only relatives: better than random, far below actual practice, mean performance only 27% of it.
- Strict **merit** (captures so far) is the best of the rules, and still about 20% below actual practice.

Figure 5: Selection effect under counterfactual promotion rules



Reading the distributions

- Only 2 out of 1,000 random draws of promotees beat the set the Admiralty actually chose.
- A machine learning rule fed only pre-promotion data does best. Actual practice reaches about three quarters of that upper bound.
- In aggregate: **407** more enemy vessels than random promotion would have produced, about 10% of all captures.
- Against strict seniority the gain is 895 ships, about 20%.

Fast-tracking and filtering

- Two ways to depart from a merit rule: promote a man the rule would skip, or hold back a man the rule would promote.
- Both raised performance, but only among the connected. For unconnected officers, departing from the rule did not help.
- **Fast-tracking** contributed more than filtering.
- Rodger: “far from clear that any method since devised has identified and promoted merit any more efficiently”.

Voth and Xu (2022) in the framework

- In the Navy, discretion worked as the use of **private information**.
- Selection: admirals read the type of their own relatives better than that of strangers, and acted on it in both directions.
- No sign of the incentive effect Xu (2018) found: officers do not fight better merely because a relative sits at the Admiralty.
- Connection and performance are complements.

Colonial Office and Royal Navy

- Same structure: discretionary appointment, connection measured by lineage, performance mapped to one aggregate outcome.
- Colonial Office: connected governors got better posts and raised less revenue. Connection substituted for performance.
- Royal Navy: connected promotees fought better, for reasons the service record does not show. Connection complemented performance.
- A possible interpretation: in the Colonial Office the connection created an **obligation** to the governor; in the Navy it gave the admirals information about the officer.

Why the two might differ

- Voth and Xu: discretion is likelier to pay where the job is a highly skilled leadership post and the paper record predicts badly.
- Their own evidence points that way: the advantage is concentrated where discretion was full.
- Xu (2018): where the connection also delivered the next posting, the connected governor had less reason to work.
- Discussion: which of the two cases do the bureaucracies you know most resemble?

Patronage now

- Britain removed patronage from governorships in 1930; the United States passed the Pendleton Act in 1883.
- In the United States, more than 8,000 senior federal posts are still filled “at the pleasure of the President”.
- Brazil, outside the papers: appointments made outside the *concurso*, and evidence that politically aligned appointees are less qualified.
- For policy, the question is whether the appointer has information a rule cannot use, and whether he has reason to use it well.